

Comprehensive Briefing: IPv4 Classful Addressing

Summary

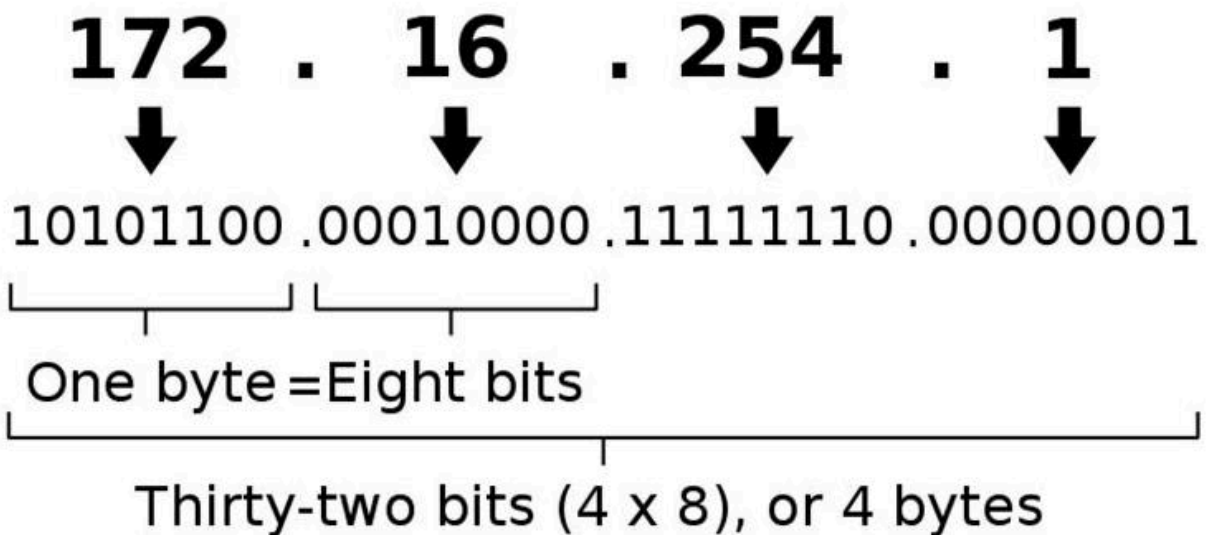
This briefing presents a **structured, exam-focused synthesis of IPv4 classful addressing**, a foundational topic in Computer Networks and a **frequent source of direct GATE questions**. IPv4 uses a **32-bit address space**, yielding **2^{32} unique addresses**, which are divided into **five predefined classes (A–E)** based on fixed leading bits.

Key technical insights include:

- The distinction between **theoretical vs practical address counts**
- The role of **Network ID (NID)** and **Host ID (HID)**
- The **all-zeros and all-ones restriction**
- The **special-purpose 127.x.x.x loopback block**
- **Fast class identification using the first octet**

1. Fundamentals of IPv4 Architecture

An IPv4 address (dotted-decimal notation)



An IPv6 address (in hexadecimal)

2001:0DB8:AC10:FE01:0000:0000:0000:0000

↓ ↓ ↓ ↓ └───┘
2001:0DB8:AC10:FE01:: Zeroes can be omitted

└───┘
10000000000000000000000000000000:0000110110111000:1010110000010000:1111111000000001:
00000000000000000000000000000000:00000000000000000000000000000000

An IPv4 address is a **32-bit binary number**, logically divided into **four 8-bit octets**.

Address Space

- **Total addresses:**
[
 $2^{32} = 4,294,967,296$
]

Representation Formats

1. **Dotted Binary**
11000000.10101000.00000001.00000001

2. Dotted Decimal

192.168.1.1 (each octet: 0–255)

3. Hexadecimal

Base-16 notation (4 bits per hex digit)

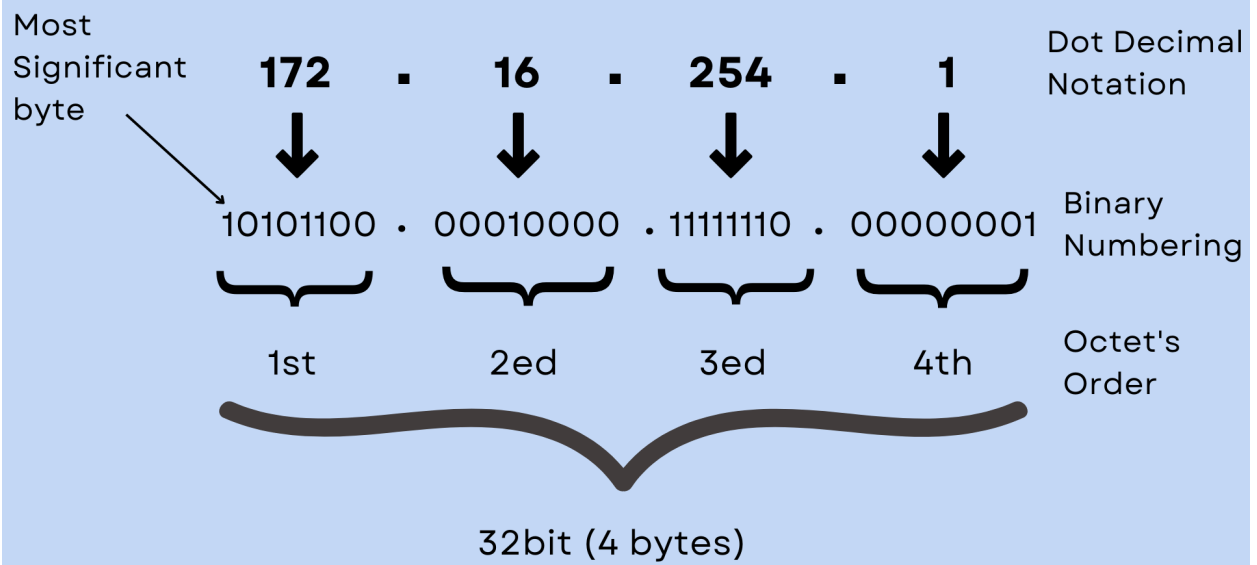
1.1 Address Space Distribution by Class

The **first few bits of the first octet** determine the class and hence the fraction of total IPv4 space:

Class	Address Space Share	Total Addresses
A	50%	2^{31}
B	25%	2^{30}
C	12.5%	2^{29}
D	6.25%	2^{28}
E	6.25%	2^{28}

This unequal distribution reflects early assumptions about organization sizes—which later proved inefficient.

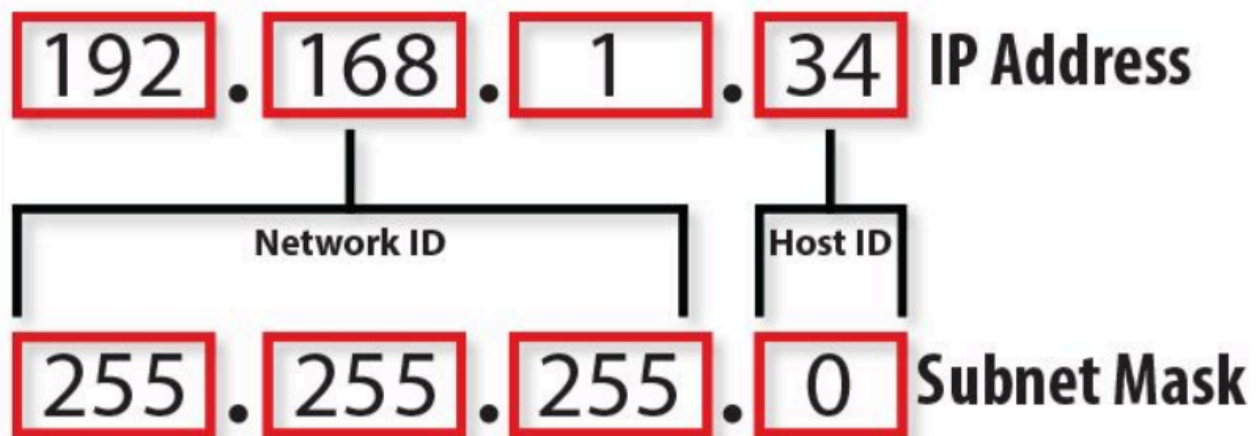
2. IPv4 Class Structure and Prefix Bits



BOORD

Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 – 127	0XXXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^7
Class B	128 – 191	10XXXXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192 – 223	110XXXXXX	192.0.0.0-223.255.255.255	255.255.255.0	2^8-2	2^{21}
Class D (Multicast)	224 – 239	1110XXXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 – 255	1111XXXXX	240.0.0.0-255.255.255.255			



Class Identification Rules

Classes	Leading Bits	NID Bits	HID Bits	1st Octet Range	Purpose
A	0	8	24	1–126	Large networks
B	10	16	16	128–191	Medium networks
C	110	24	8	192–223	Small networks
D	1110	N/A	N/A	224–239	Multicasting
E	1111	N/A	N/A	240–255	Research / Reserved

Key Insight:

Classes D and E **do not use NID/HID division** and are never assigned to hosts.

3. Reserved and Special-Purpose Addresses

3.1 All-Zeros and All-Ones Rule

An IP address is **not assignable to a host** if:

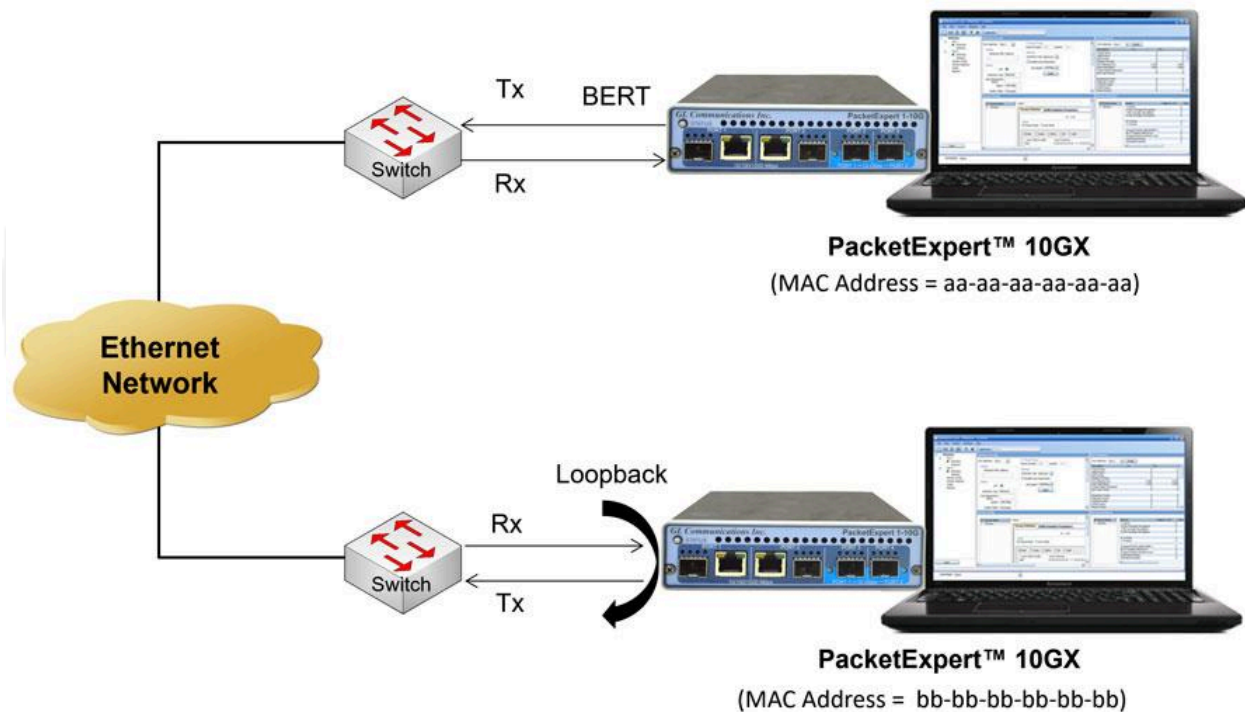
- **Network ID** is:
 - All zeros
 - All ones

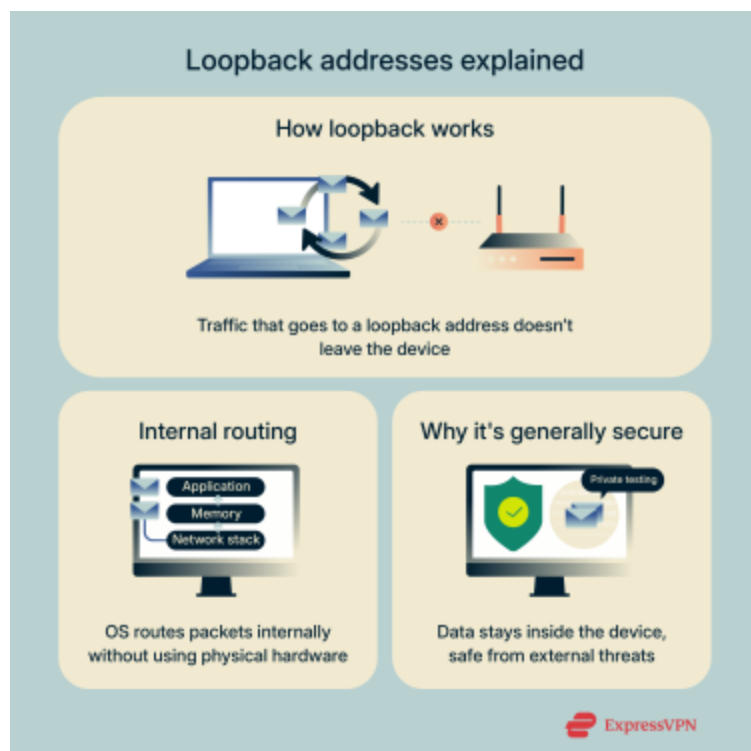
- **Host ID is:**
 - All zeros → **Network Address**
 - All ones → **Directed Broadcast Address**

These exclusions explain why practical host counts always use:

$$[2^n - 2]$$

3.2 Loopback Addresses (127.x.x.x)





```
MacBook-Air-4: ~ -- -bash -- 80x24
Last login: Mon Sep 29 17:30:36 on ttys000
You have new mail.

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
MacBook-Air-4:~ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1): 56 data bytes
64 bytes from 127.0.0.1: 14 packets transmitted, 4 packets received, 0.0% packet loss
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.149 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.234 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.215 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.149/0.192/0.234/0.034 ms
MacBook-Air-4:~ root$
```

The **entire 127.x.x.x block** is reserved for **loopback testing**.

Purpose

- Test TCP/IP stack
- Verify NIC functionality

- Enable **inter-process communication (IPC)**

Constraints

- Cannot be used as **source IP**
- Used only as **destination IP**
- Packet never leaves the host

Practical Loopback Addresses

- Address space: **127.0.0.0 – 127.255.255.255**
- Excluding:
 - **127.0.0.0** (all-zero host)
 - **127.255.255.255** (all-one host)

[
 $\text{Usable loopback IPs} = 2^{24} - 2$
]

4. Practical Network and Host Calculations

4.1 Class A

- **NID bits:** 8 (1 fixed as 0)
- Reserved networks:
 - **0.x.x.x**
 - **127.x.x.x**

Practical Networks

[
 $2^{8-1} - 2 = 126$
]

Hosts per Network

[
 $2^{24} - 2 = 16,777,214$
]

4.2 Class B

- **NID bits:** 16 (2 fixed as 10)

Networks

[
 $2^{\{16-2\}} = 2^{\{14\}} = 16,384$
]

Hosts per Network

[
 $2^{\{16\}} - 2 = 65,534$
]

4.3 Class C

- **NID bits:** 24 (3 fixed as 110)

Networks

[
 $2^{\{24-3\}} = 2^{\{21\}}$
]

Hosts per Network

[
 $2^{\{8\}} - 2 = 254$
]

5. Loopback-Based Network Troubleshooting

When **Host A cannot reach Host B**, possible failure points include:

1. Host B is down
2. Physical link failure
3. Router malfunction
4. Local NIC or TCP/IP stack issue

Diagnostic Strategy

- Send packets to **127.x.x.x**
- If loopback fails → **local problem**
- If loopback succeeds → issue lies **outside the host**

This makes loopback testing the **first step in network diagnostics**.

6. Rapid Class Identification (GATE Shortcut)

First Octet	IP Class
-------------	----------

1 – 126	Class A
---------	---------

127	Loopback (Reserved)
------------	------------------------

128 – 191	Class B
-----------	---------

192 – 223	Class C
-----------	---------

224 – 239	Class D
-----------	---------

240 – 255	Class E
-----------	---------